"Experience of Global Education..... World Impact "

GREEN PEACE LINCOLN COLLEGE

(Affiliated to Lincoln University College, Malaysia)
Itahari, Sunsari, Nepal

FACULTY OF MBA/BBA/BIT

ASSIGNMENT COVER PAGE

Please fill in all the required details for your assignment to be accepted.

Student's Name	Sahil Chaudhary			
Year/Semester	2nd/ 3rd		**2024**	
Program	BIT			
Subject Name/ Code	E-Commerce/BIT232			
LC Number	LC00023000959			
Lecturer's Name	Nabin niroula			
Assignment Tittle (1) ☐ (2) ☐	Strategies for implementing security for E-commerce			
No. of Page (excluding this page)	6			
Required Words	1000-1500		1159	
Soft Copy Included		**Yes** ✓	**/ No** ☐	

DECLARATION BY STUDENTS:

I Certify that assignment is my own work in my own words. All resources have been acknowledged and the content has not been previously submitted for assessment to ***LINCOLN*** *or elsewhere. I also confirm that I have kept a copy of this assignment.*

Signed:————————

Date: 1 5 0 1 2 0 2 6

1. Introduction

E-Commerce refers to the process of buying and selling goods and services through electronic networks, mainly the internet. In the modern digital era, E-Commerce has become an essential part of business operations because it allows organizations to reach global customers efficiently and at lower operational costs. Online platforms are now widely used for product promotion, order processing, online payments, and customer service management.

With the rapid growth of E-Commerce, the amount of sensitive information handled by these systems has also increased significantly. E-Commerce systems store and process personal customer data, usernames, passwords, credit card information, and transaction records. Due to the high value and sensitivity of this information, security has become a major concern (Network, 2023). Security threats such as hacking, identity theft, online fraud, and data breaches can cause financial loss and damage the reputation of both customers and businesses. Therefore, implementing effective security strategies is essential to protect digital assets, ensure transaction safety, and maintain customer trust in E-Commerce systems.

1.1 Statement of the Problem

The rapid expansion of E-Commerce has increased the dependency of businesses on online platforms for sales, communication, and financial transactions. While this growth provides convenience and efficiency, it also exposes systems to various security risks. Unauthorized access, data manipulation, malware attacks, and online fraud are common threats faced by E-Commerce platforms.

Many organizations fail to implement proper security measures due to lack of awareness, technical limitations, or high implementation costs. As a result, security breaches may lead to financial losses, misuse of customer data, legal issues, and loss of customer confidence. Therefore, there is a serious need to understand and apply effective security strategies to protect E-Commerce systems and ensure safe online transactions.

1.2 Objectives of the Study

- To understand the importance of security in E-Commerce systems.
- To study different strategies used to protect intellectual property in E-Commerce.
- To develop basic knowledge of legal and technical security mechanisms used in E-Commerce.

1.3 Significance of the Study

This study is significant because it helps students and beginners understand the importance of security in E-Commerce environments. With the increasing number of online users, security threats such as hacking, identity theft, and online fraud are also rising. This study explains various security strategies that help protect customer data, business information, and online transactions.

The study also provides knowledge about legal protections such as copyright, patents, and trademarks, along with technical measures like encryption and firewalls. Understanding these concepts helps learners prepare for real-world E-Commerce applications and promotes safe online business practices.

1.4 Limitations of the Study

- The study is based only on theoretical concepts.
- Real-time case studies and practical implementation are not included.
- The study focuses on basic security strategies rather than advanced technologies.
- Information is collected from secondary sources only.

2. Methodology and Data Collection

The methodology used in this study is based on descriptive and theoretical analysis. The required data has been collected from secondary sources such as textbooks, academic journals, reference books, course materials, and reliable online resources related to E-Commerce security.

The collected information was carefully reviewed, analyzed, and organized according to the assignment requirements. This method helped present the content in a clear, systematic, and easy-to-understand manner without conducting surveys or experiments.

3. Results and Discussion

3.1 Intellectual Property Protection in E-Commerce

Intellectual Property (IP) protection is a major concern in E-Commerce because digital content can be easily copied and distributed without permission. IP includes software, website content, images, logos, and innovative technologies. Legal frameworks such as copyright, patent, and trademark laws help protect these digital assets ((GIPC), 2024).

Copyright protects original works like website text, software code, and multimedia content. Trademarks protect brand identity and prevent fake websites from misusing brand names. Patents protect new technologies and innovative systems used in E-Commerce ((WIPO), 2023). However, enforcing these laws is challenging due to the global nature of the internet.

3.2 Methods of Protecting Intellectual Digital Work

Software Metering:
Software metering controls how software is used by limiting access based on time, number of users, or license conditions. It helps prevent illegal copying and unauthorized usage.
Digital Watermarking:
Digital watermarking involves embedding hidden information into digital files such as images and videos. This helps identify ownership and trace unauthorized distribution of content.

Digital Envelopes:
A digital envelope uses encryption techniques to protect sensitive information. The data is encrypted, and the encryption key is securely shared only with authorized users, ensuring data confidentiality.

3.3 Transaction Integrity Overview

Transaction integrity ensures that online transactions are completed accurately without any alteration during transmission. It guarantees that payment amounts, order details, and customer information remain unchanged. Technologies such as encryption, hash functions, and digital signatures help maintain transaction integrity. Secure protocols like HTTPS ensure safe communication between customers and E-Commerce servers (Corporation, 2023).

3.4 Protecting Electronic Commerce Assets

E-Commerce assets include hardware, software, databases, and network infrastructure. Protecting these assets is essential for smooth business operations.

The CIA Triad is commonly used to secure E-Commerce assets:

- Confidentiality: Ensures that sensitive data is accessible only to authorized users.
- Integrity: Ensures that data is accurate and not modified without authorization.
- Availability: Ensures that systems and services are available to users whenever required.

3.5 Protecting E-Commerce Servers

E-Commerce servers store critical business and customer data, making them primary targets for cyberattacks (Cisco Systems, 2024).

Access Control and Authentication:
Access control restricts system access based on user roles. Authentication methods such as passwords, biometrics, and multi-factor authentication verify user identity.

Operating System Security:
Regular updates, security patches, and removal of unnecessary services help protect the server operating system from vulnerabilities.

Firewall:
Firewalls monitor network traffic and block unauthorized access, protecting E-Commerce servers from external threats.

4. Conclusion

In conclusion, security is a crucial element of E-Commerce systems. The protection of intellectual property, transaction integrity, digital assets, and servers plays a vital role in ensuring safe and reliable online business operations. A combination of legal measures, technical tools, and organizational policies is required to effectively manage E-Commerce security.

Businesses must continuously update their security strategies to deal with emerging threats. By implementing strong security mechanisms, organizations can protect customer data, build trust, and achieve long-term success in the competitive digital marketplace.

5. References

(GIPC), G. I. (2024). *Why Intellectual Property Is Important for E-Commerce (example title — adapted from content)*. Retrieved from (Global Intellectual Property Center): https://www.gipc.org/importance-of-ip-in-ecommerce

(WIPO), W. I. (2023). *Intellectual Property (IP) and E-Commerce*. Retrieved from wipo.int: https://www.wipo.int/ecommerce/en/

Cisco Systems, I. (2024). *What Is Server Security? – Threats, Best Practices, and How to Protect Your Servers*. Retrieved from Cisco: https://www.cisco.com/c/en/us/solutions/collateral/enterprise-networks/server-security/guide-c07-744955.html

Corporation, I. (2023). *What Is Transaction Security?* Retrieved from IBM: https://www.ibm.com/topics/transaction-security

Network, G. C. (2023). *E-commerce Security Threats and How to Prevent Them*. Retrieved from Global Cybersecurity Network: https://globalcybersecuritynetwork.com/blog/ecommerce-security-threats/